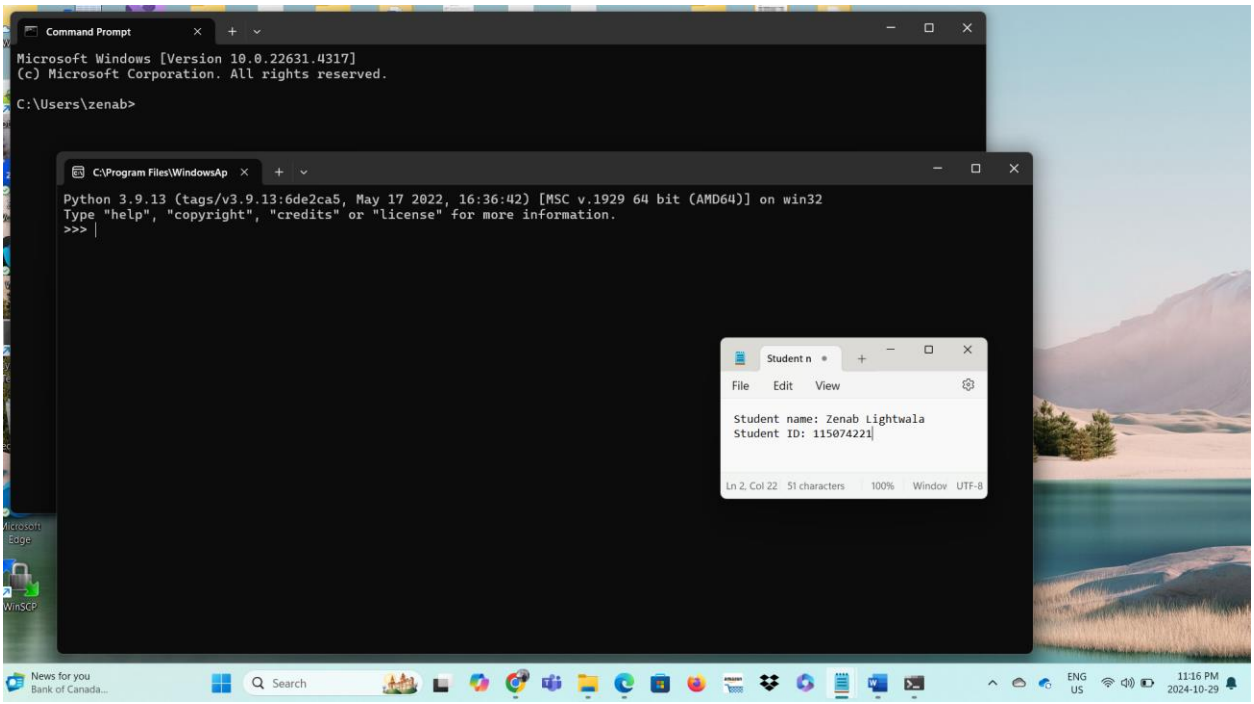
CYT180ZAA – MIDTERM HOME PROJECT

Name: Zenab Lightwala

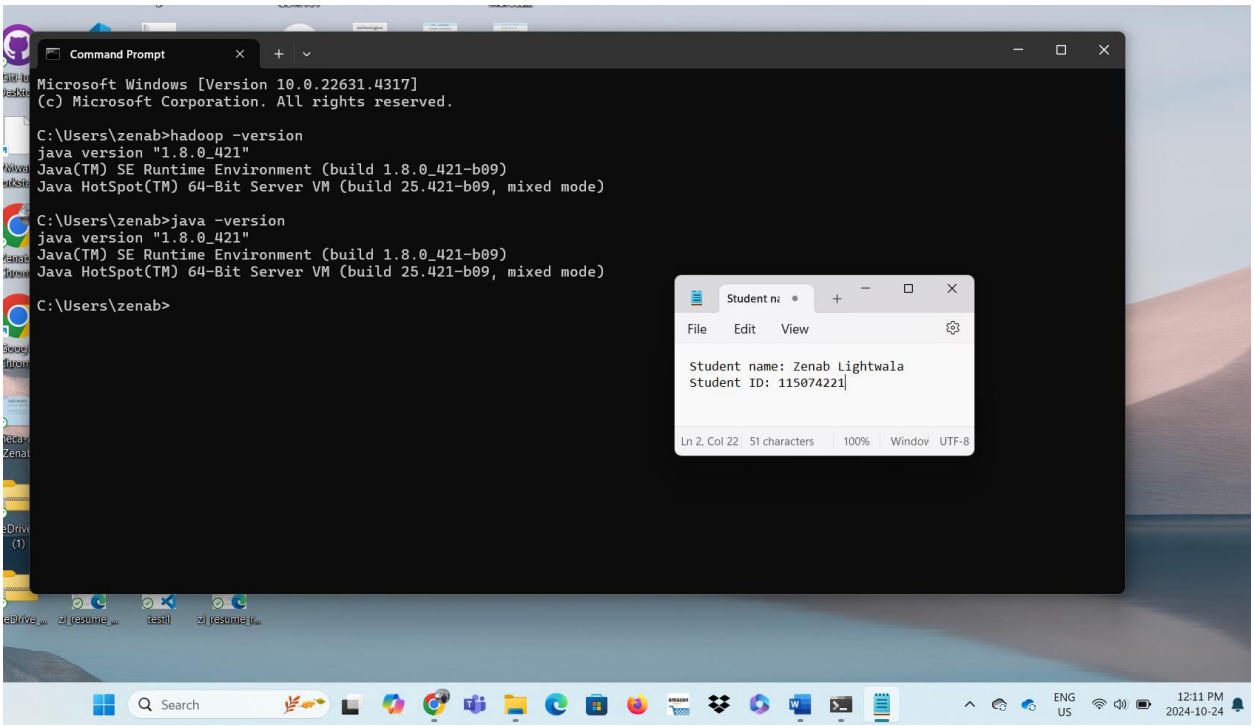
Student ID: 115074221

Professor Name: Tatiana Outkina

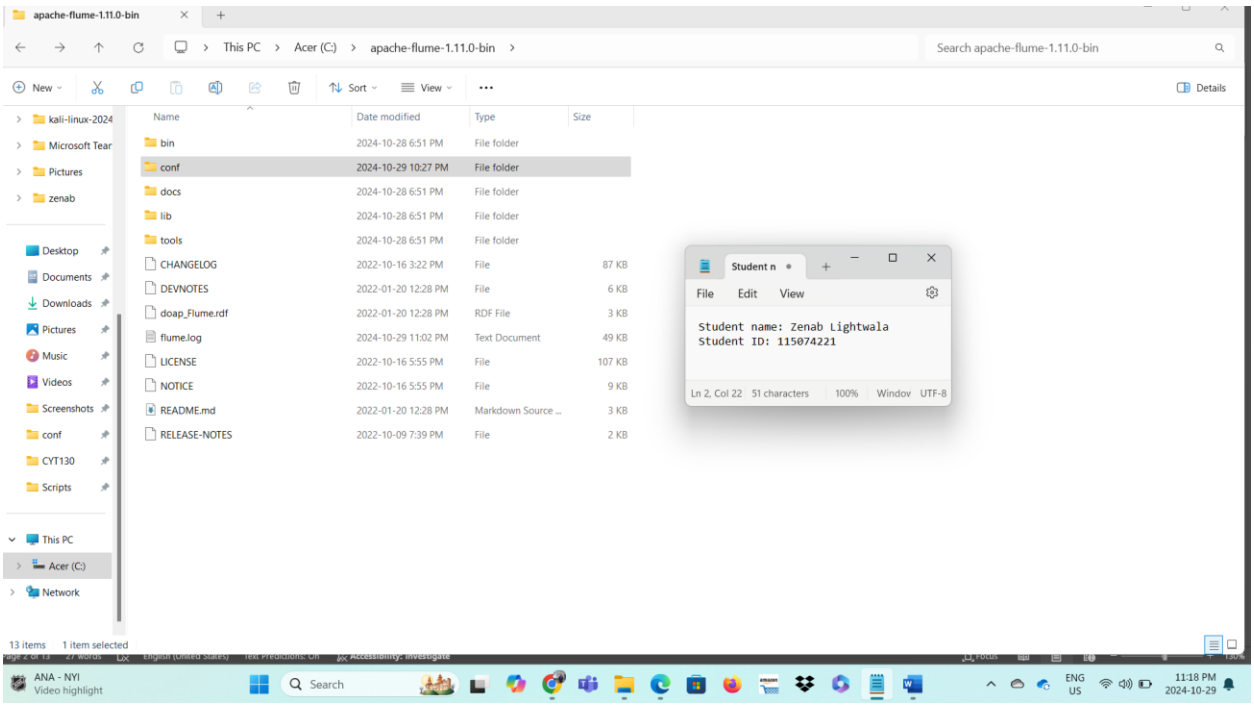
Project 1. Learn the Flume component of Hadoop Ecosystem



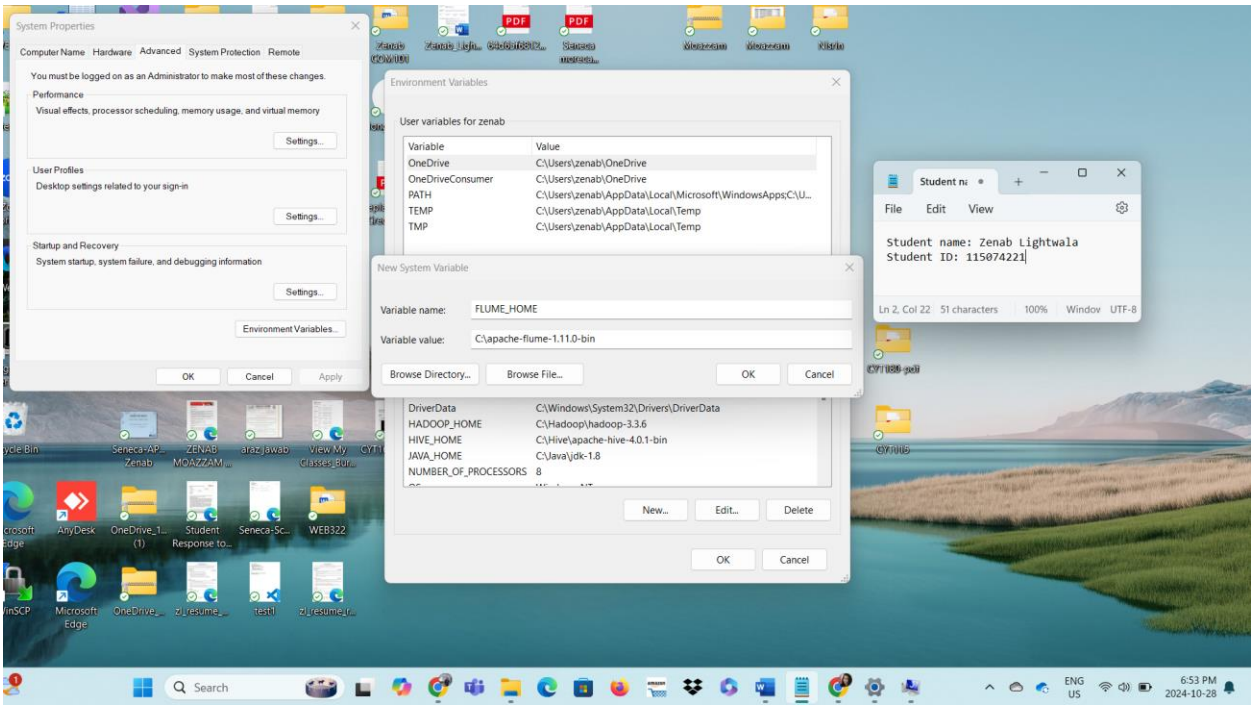
- 0 screen to show work done on my computer



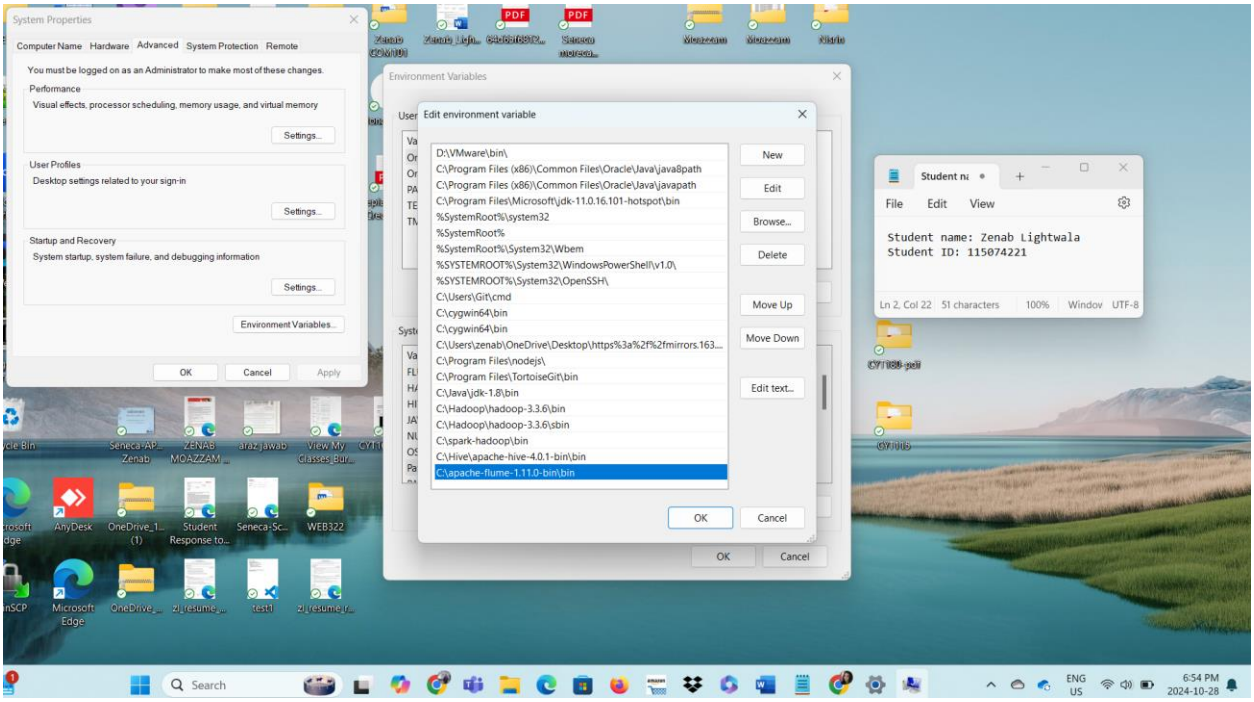
- Checking Java and Hadoop version



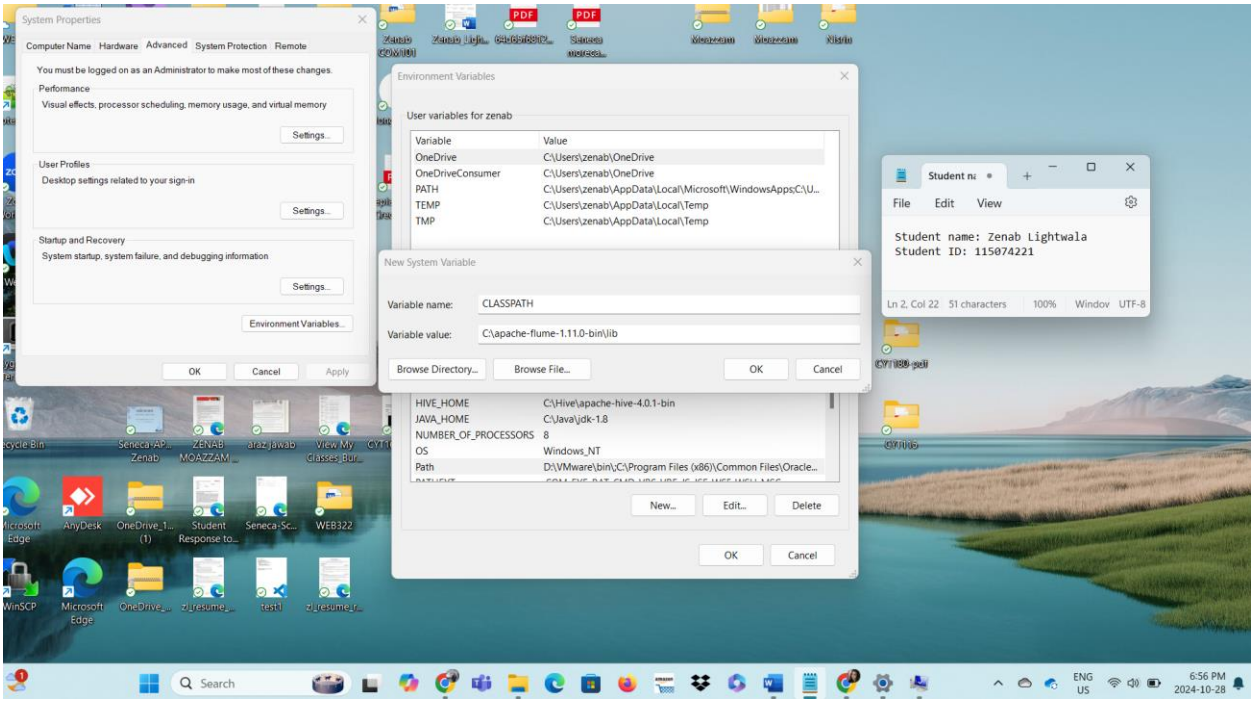
- Downloaded and installed Apache flume and moved it to C:\ drive



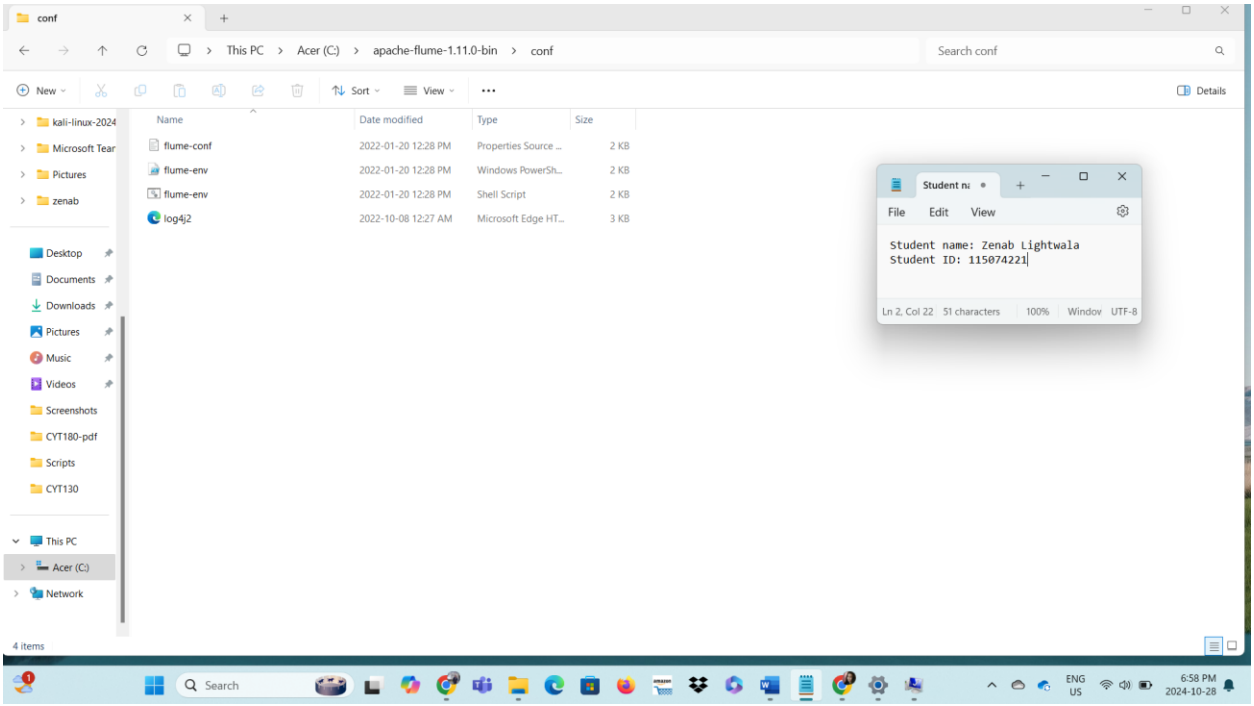
- Setting up environment variables – FLUME_HOME



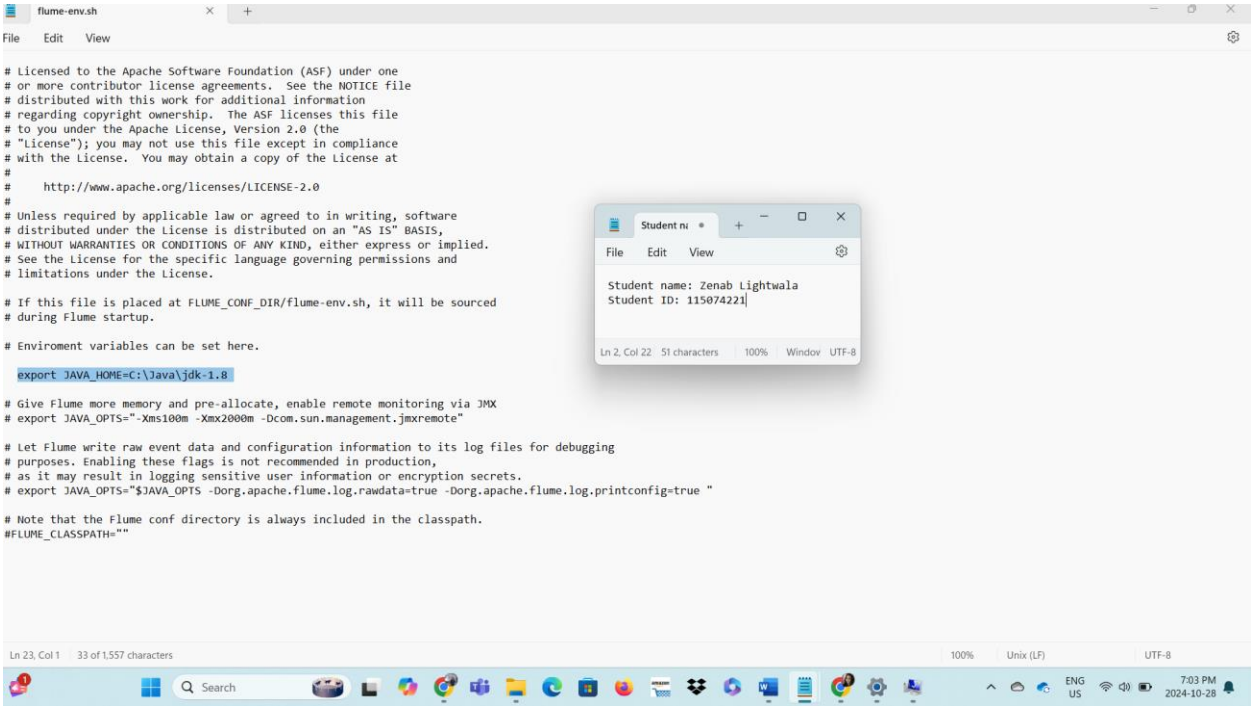
- Setting up PATH for apache flume bin directory – C:\apache-flume-1.11.0-bin\bin



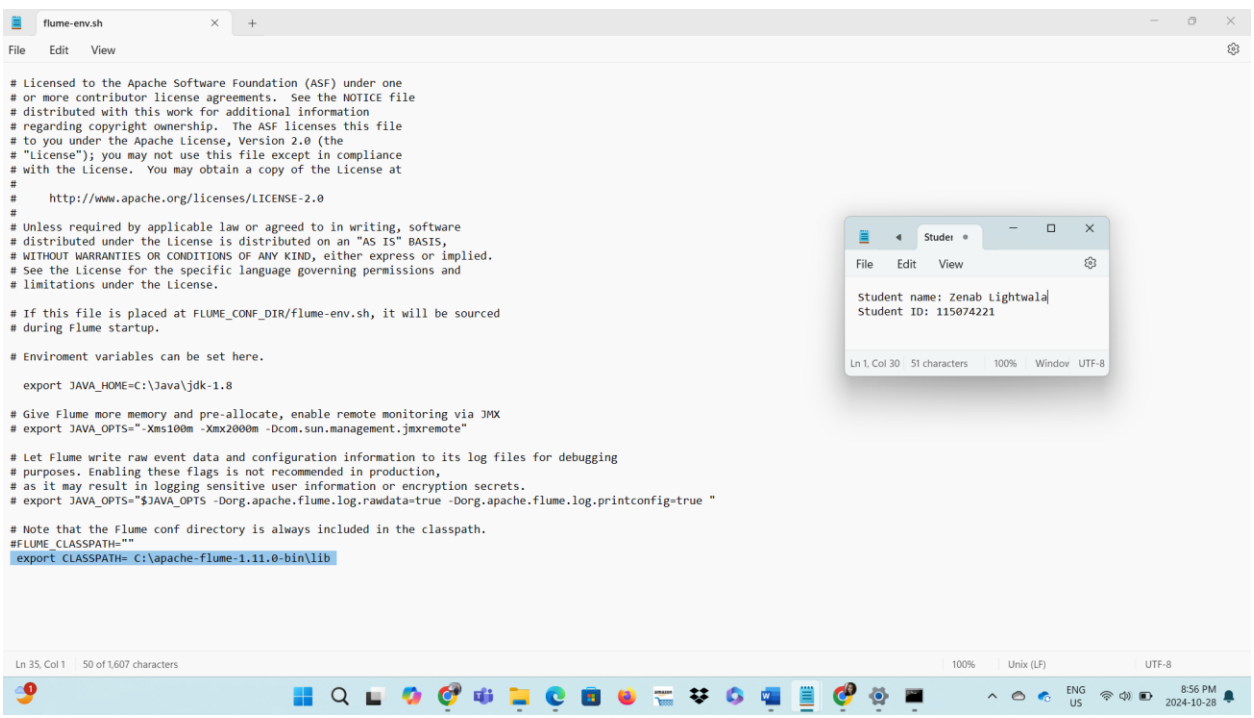
- Setting up system variable – CLASSPATH with the value – C:\apache-flume-1.11.0-bin\lib



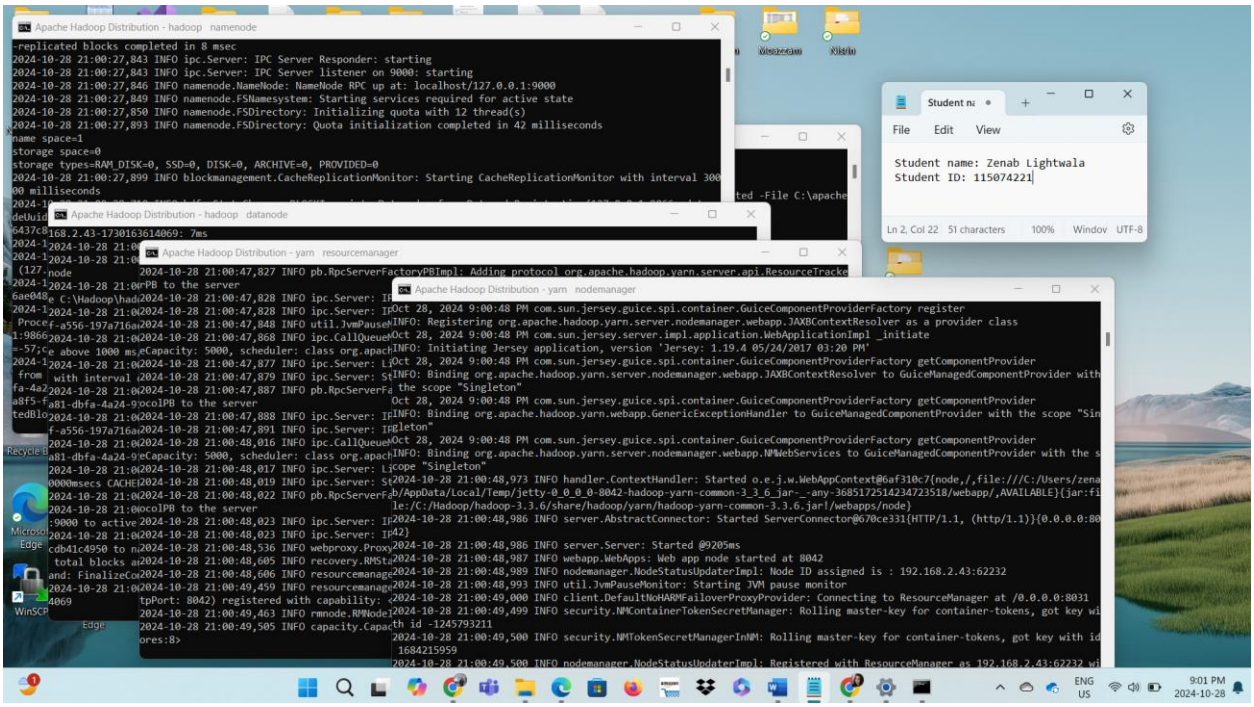
- Removed template extension



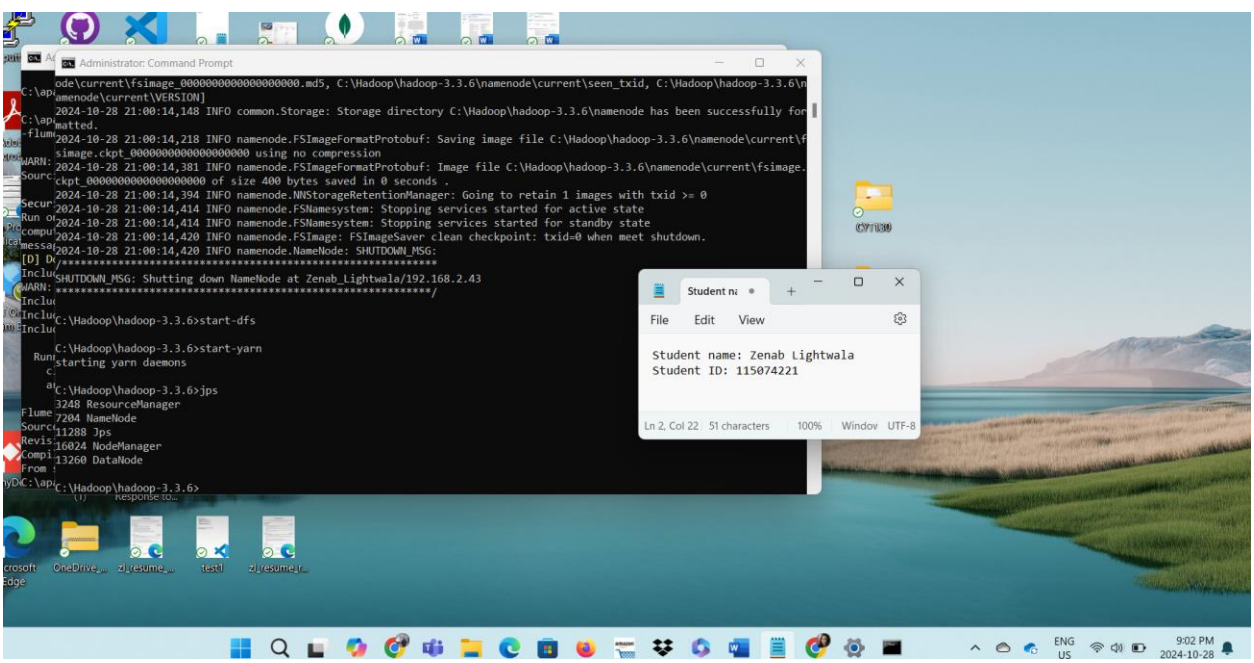
- Edited JAVA_HOME in flume-env.sh file to jdk-1.8



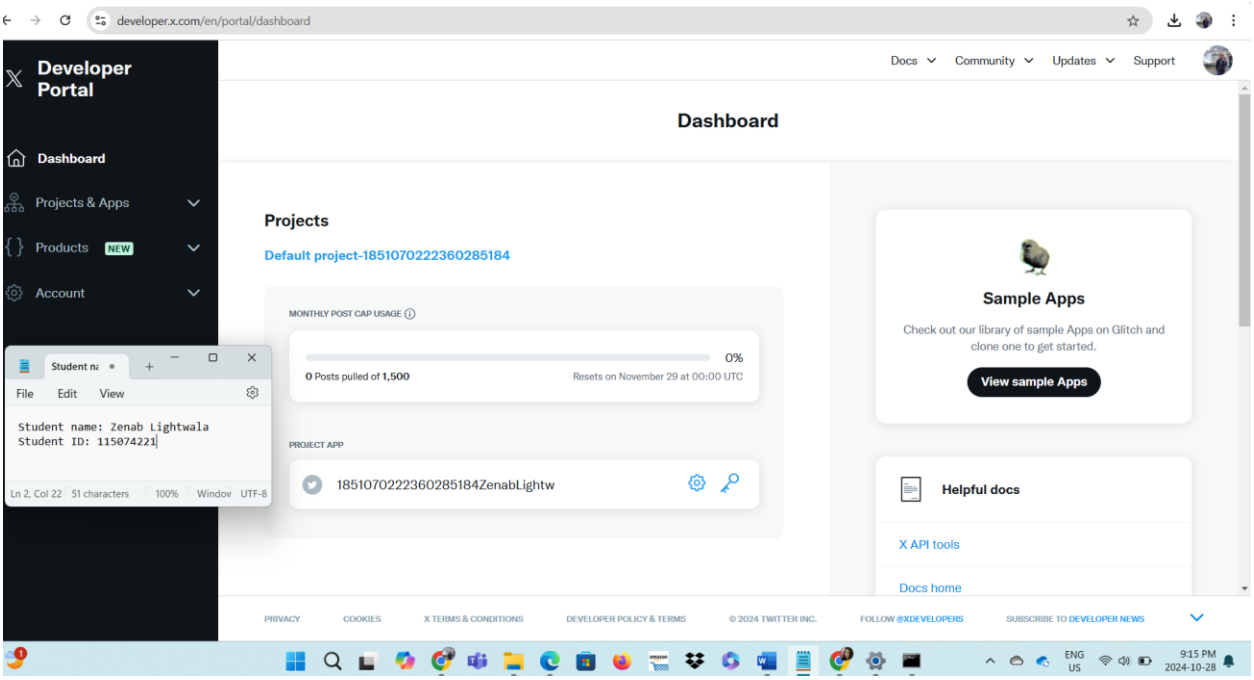
- Edited CLASSPATH to C:\apache-flume-1.11.0-bin\lib



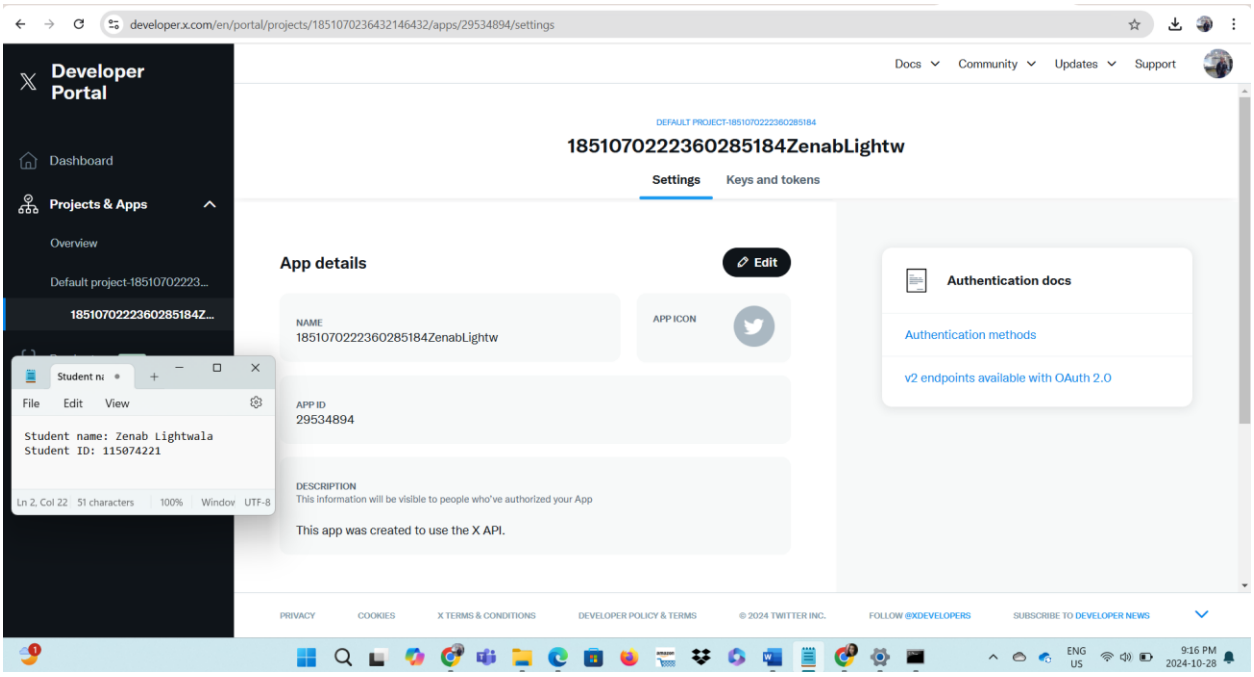
- Starting dfs and yarn nodes



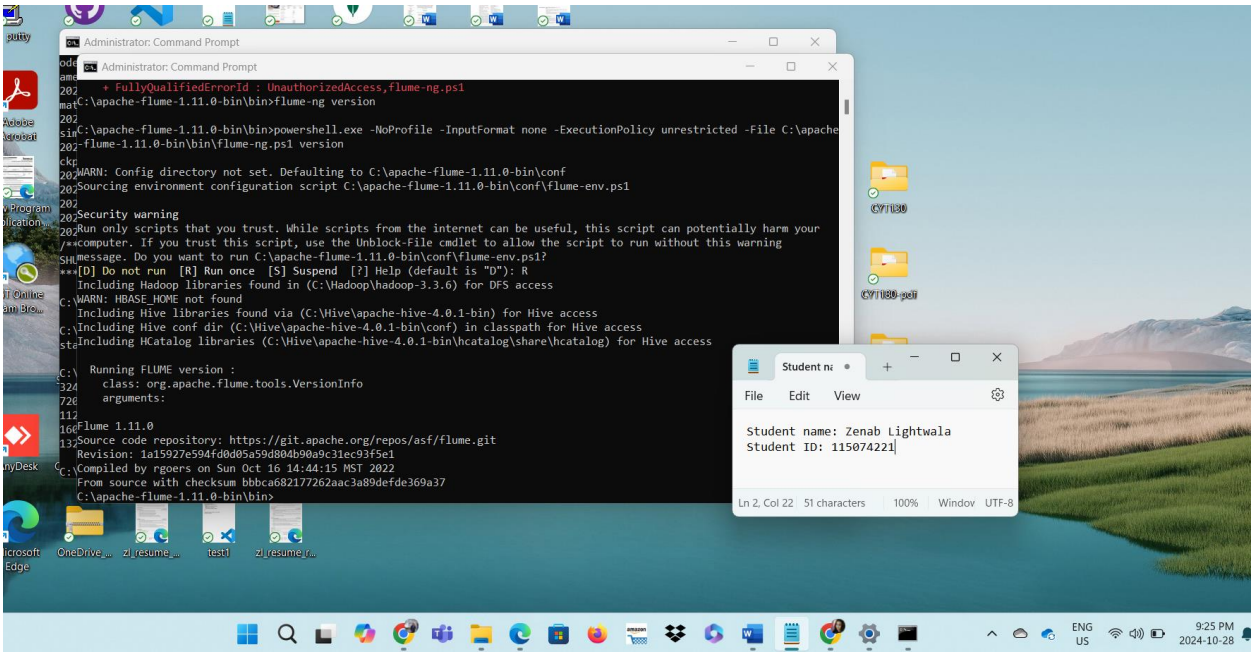
- Running jps command to confirm that all nodes are working



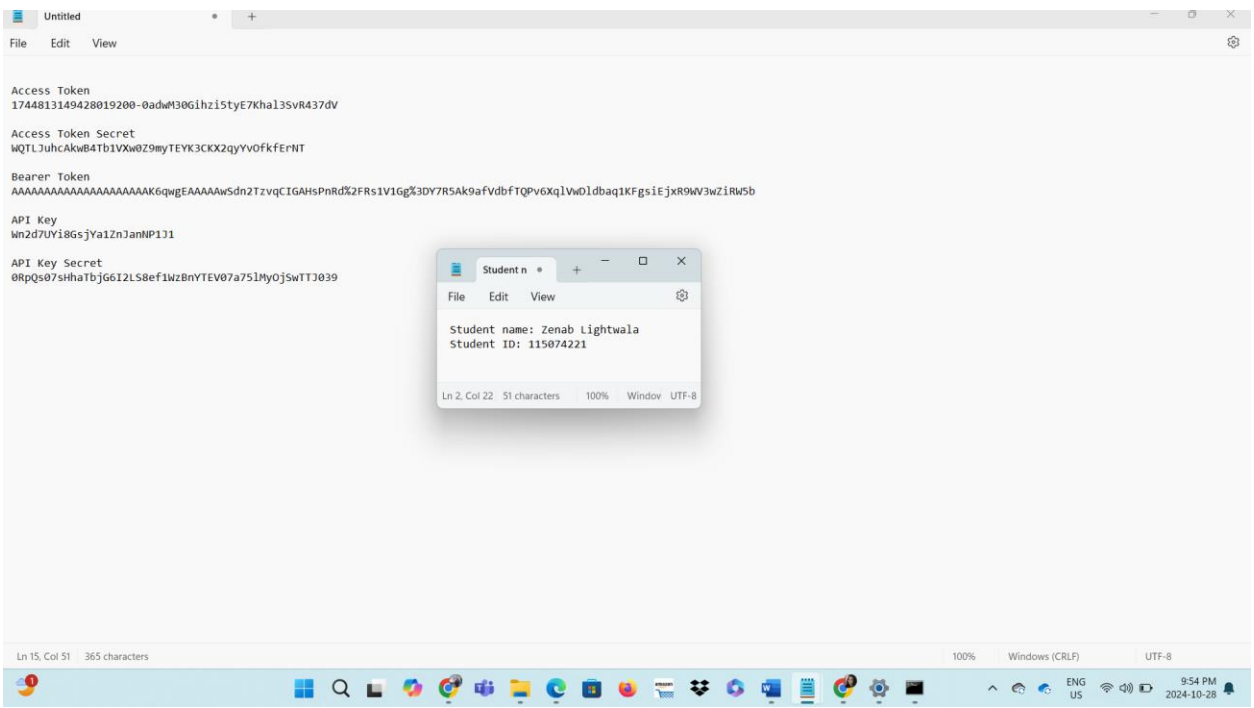
- Created free account on twitter developer site



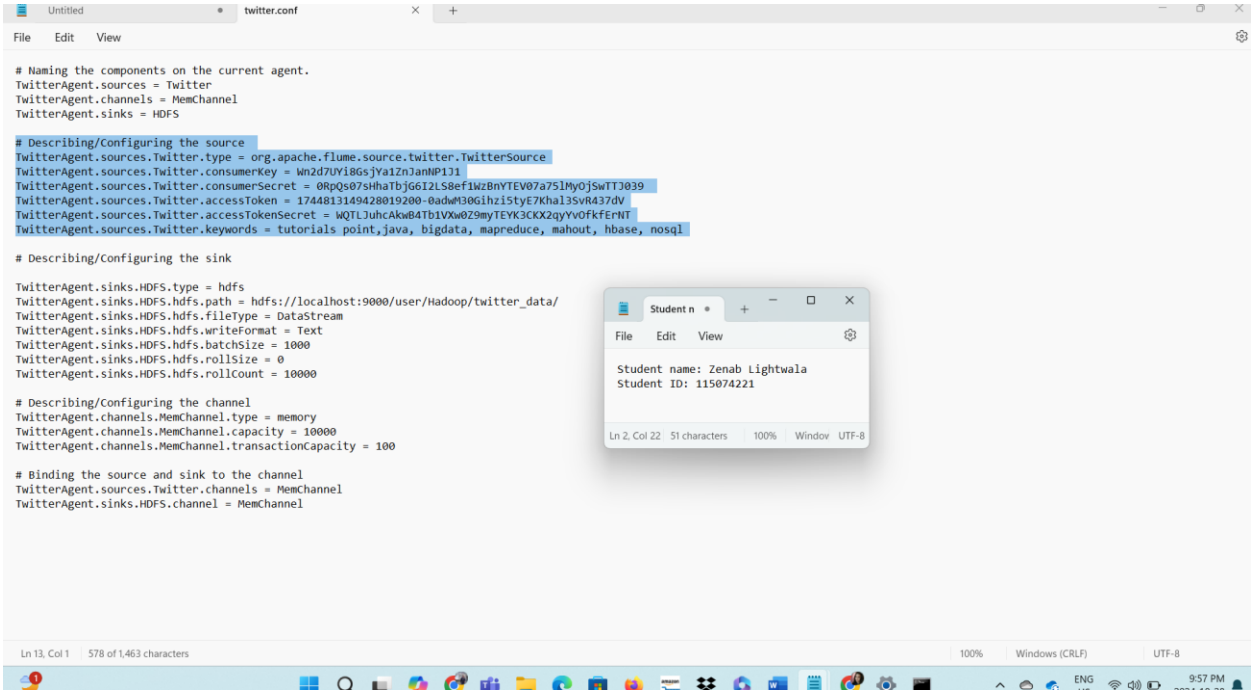
- Using the free default project



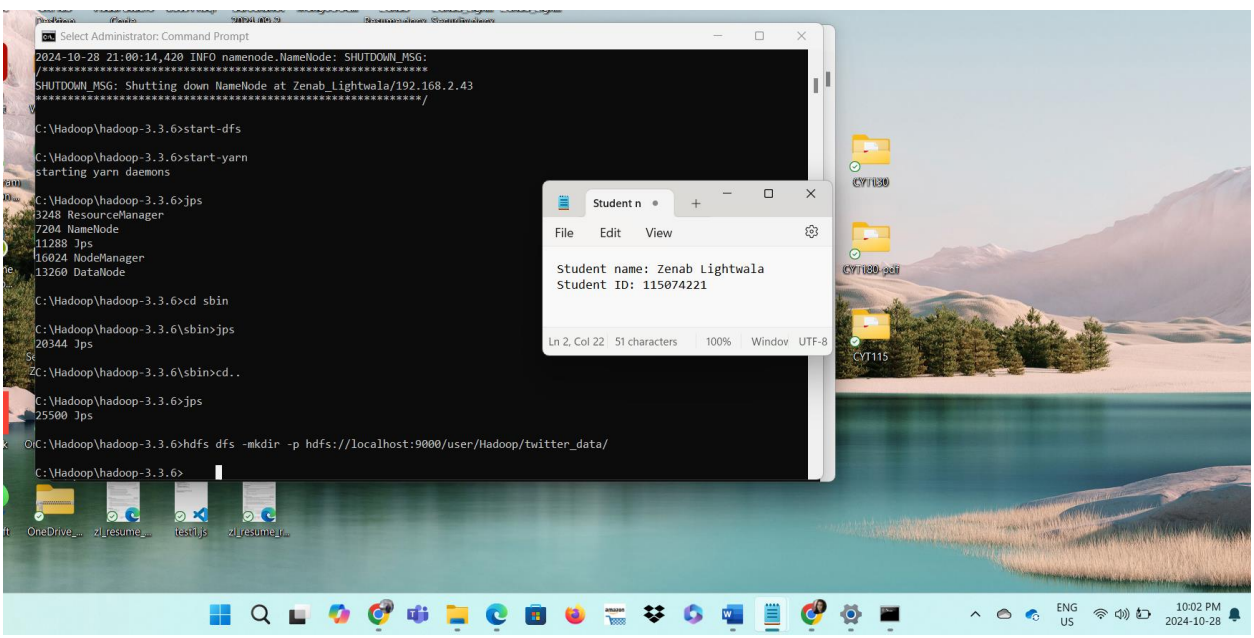
- Checking flume version



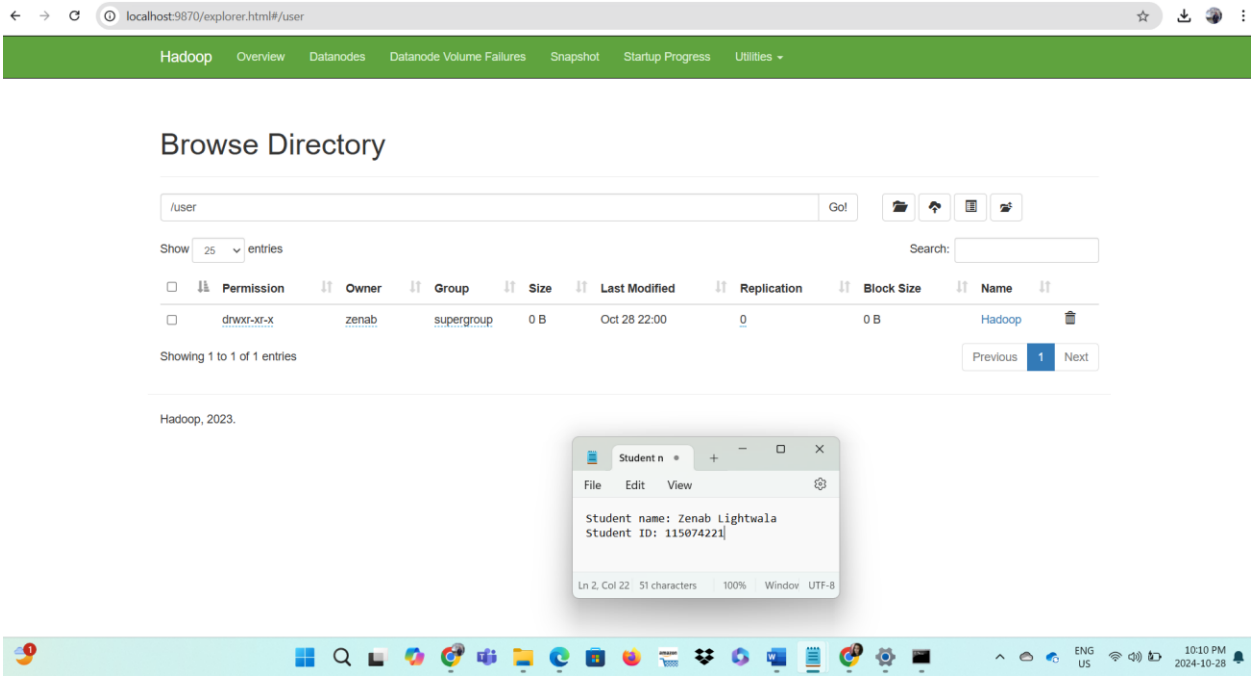
- Keys and Tokens

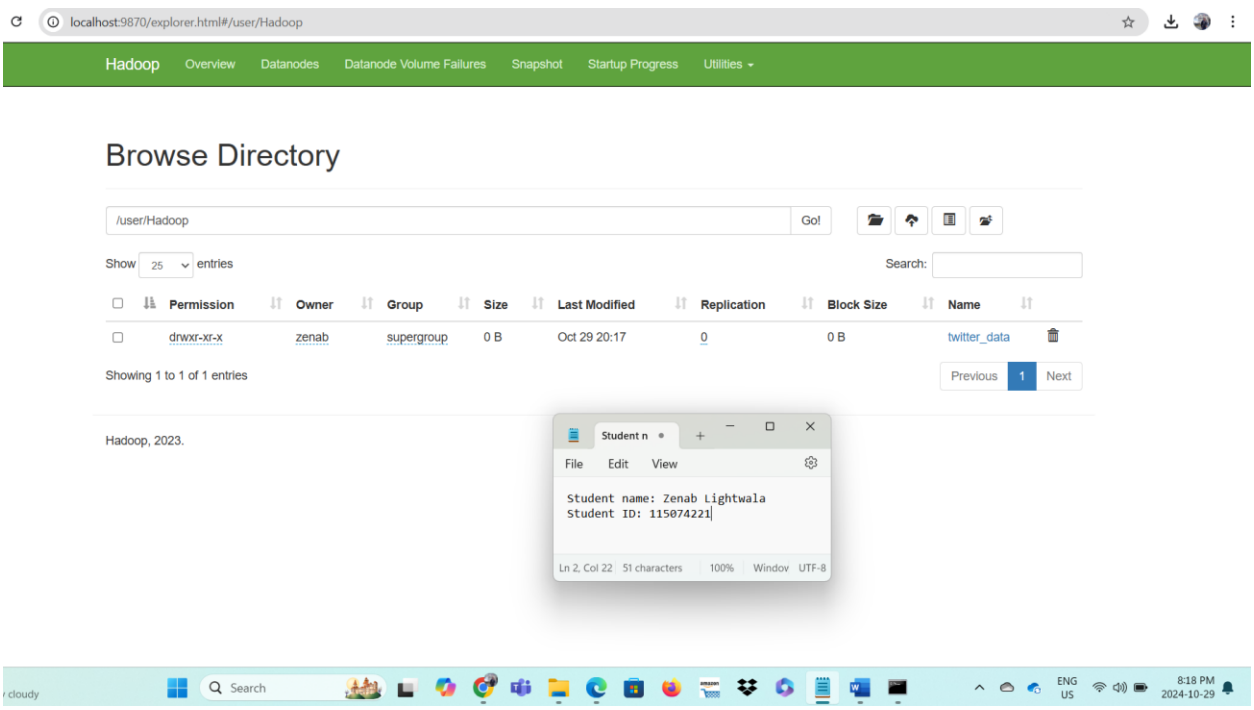


- Created a twitter.conf file in conf folder and making changes with the keys and token generated

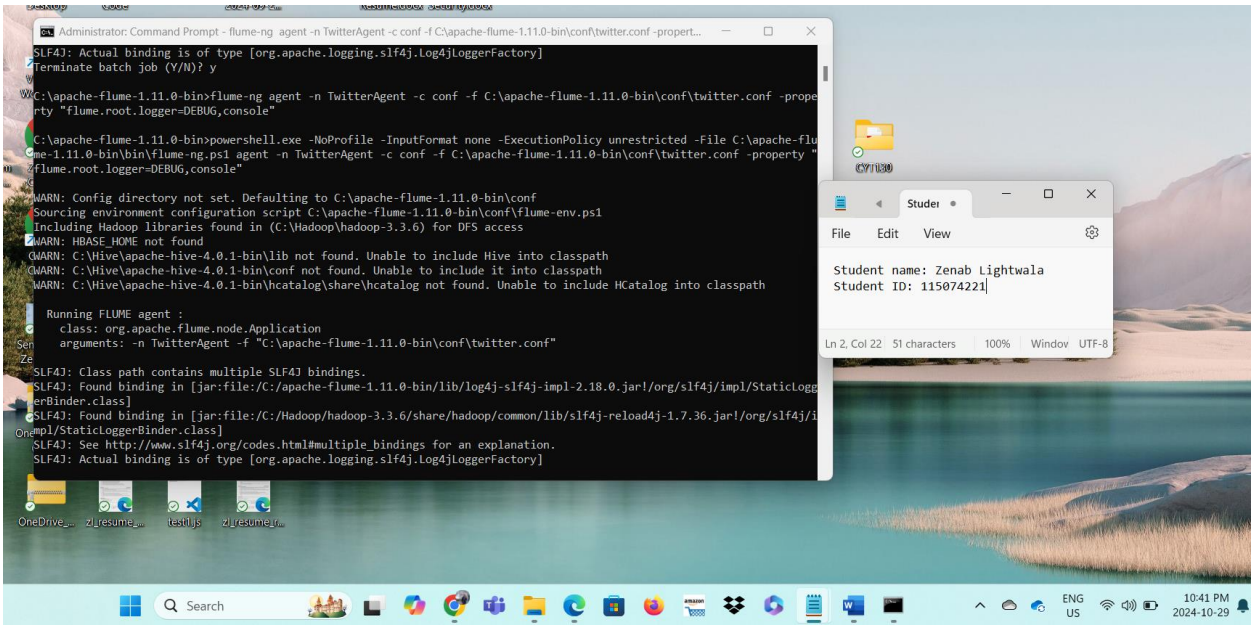


- Making a directory in user/Hadoop/twitter_data/ on localhost:9000 for us which is available on localhost:9870



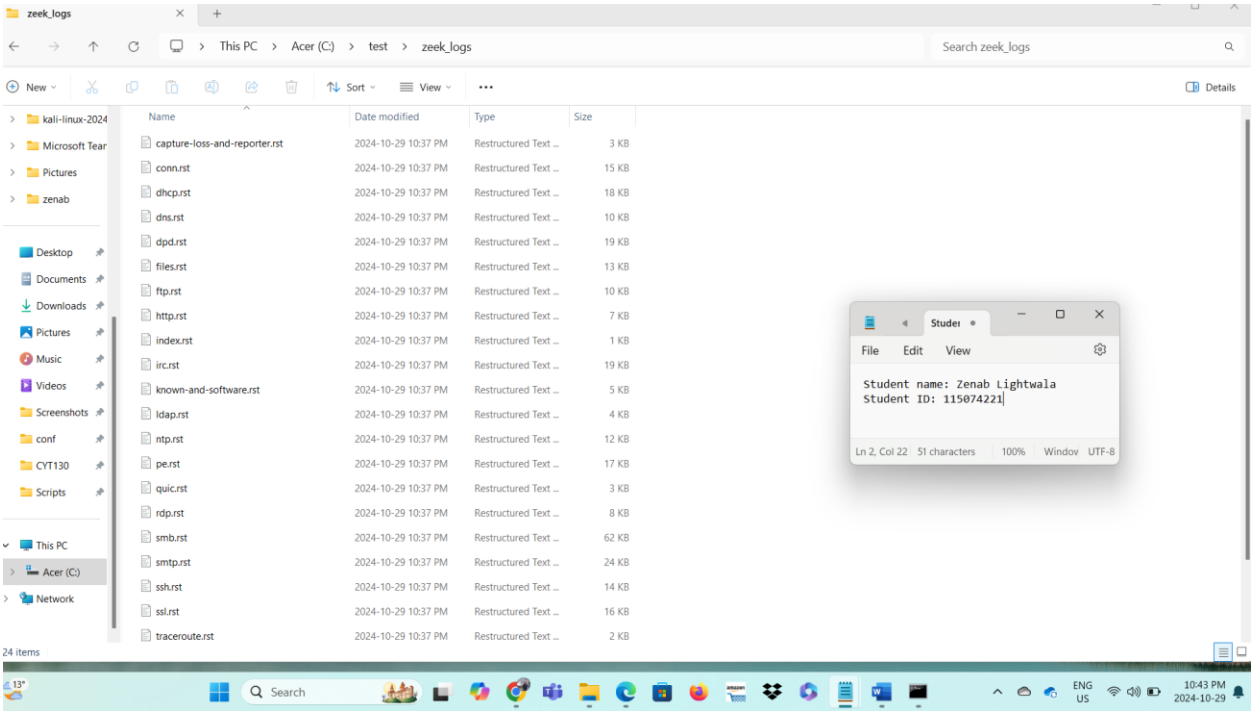


- Verifying on localhost:9870

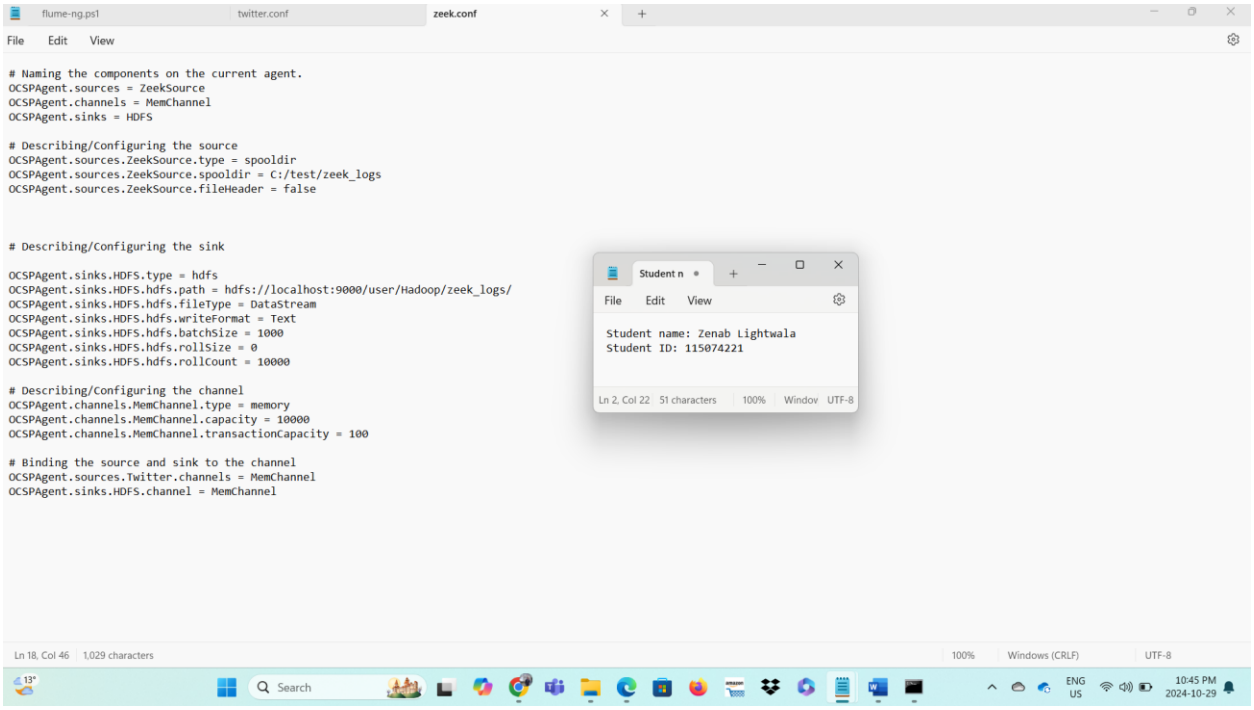


- Running the twitter.conf with command ‘flume-ng agent -n TwitterAgent -c conf C:/apache-flume-1.11.0-bin/conf/twitter.cong -property “flume.root.logger=DEBUG.console’

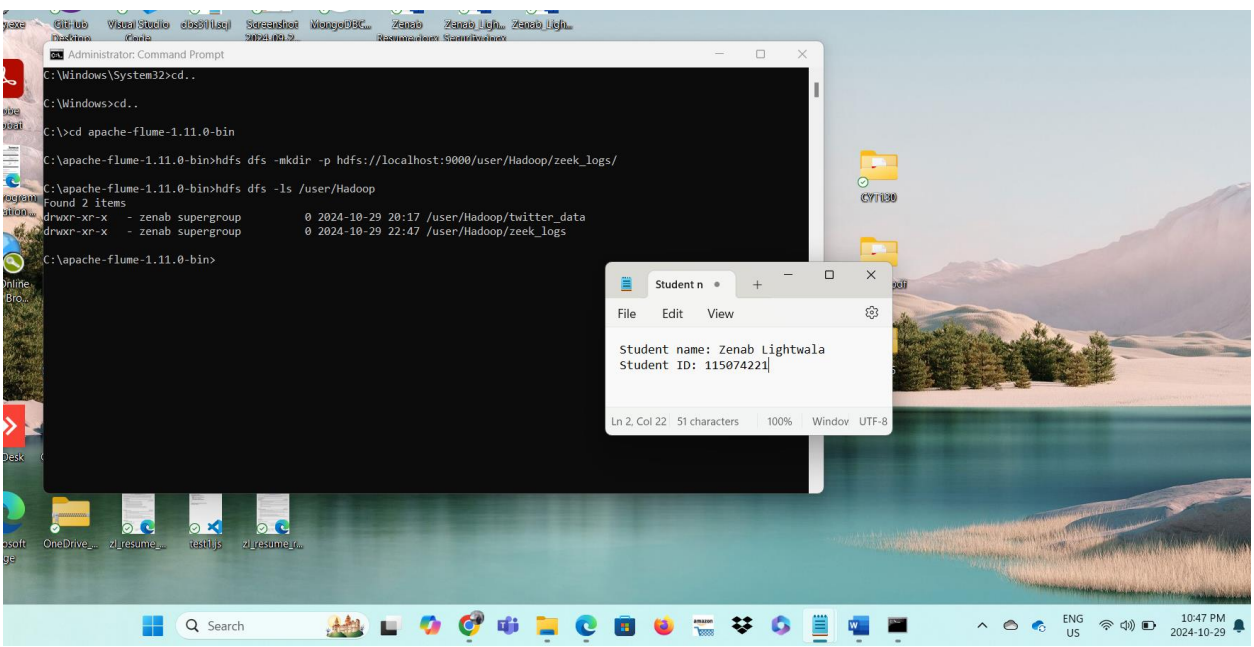
Log Files – ZEEK STORE LOG



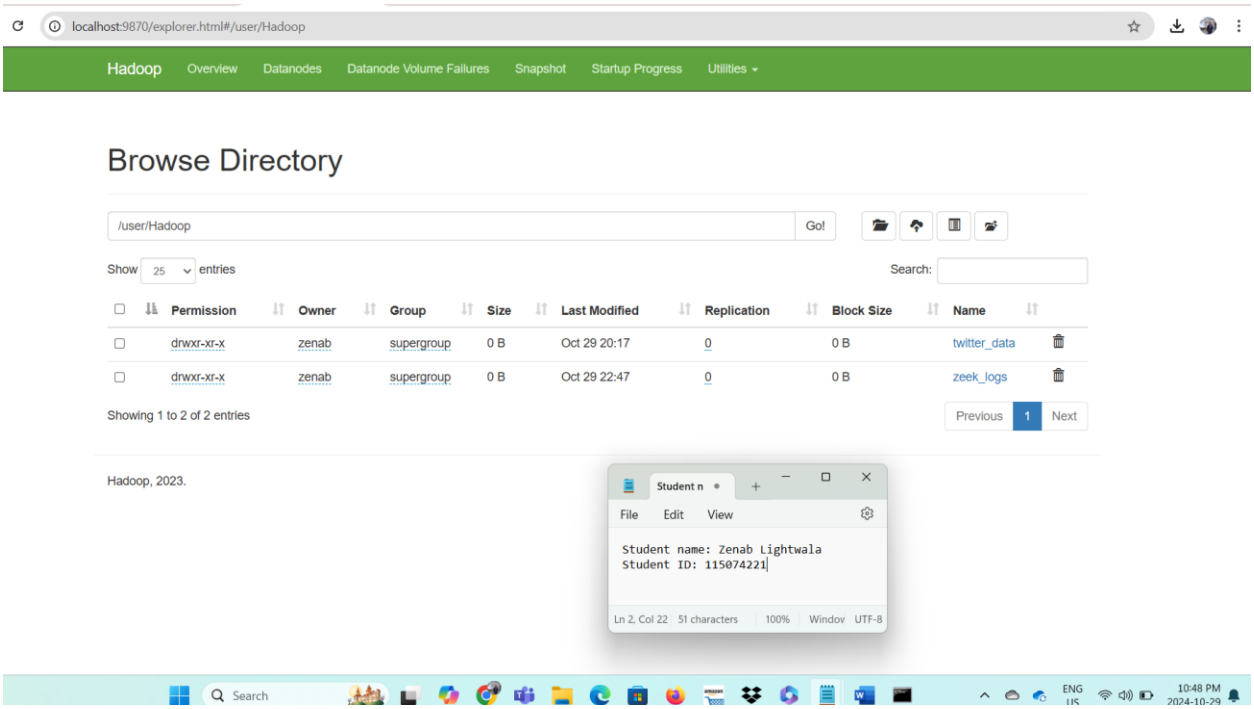
- Zeek_logs in test folder



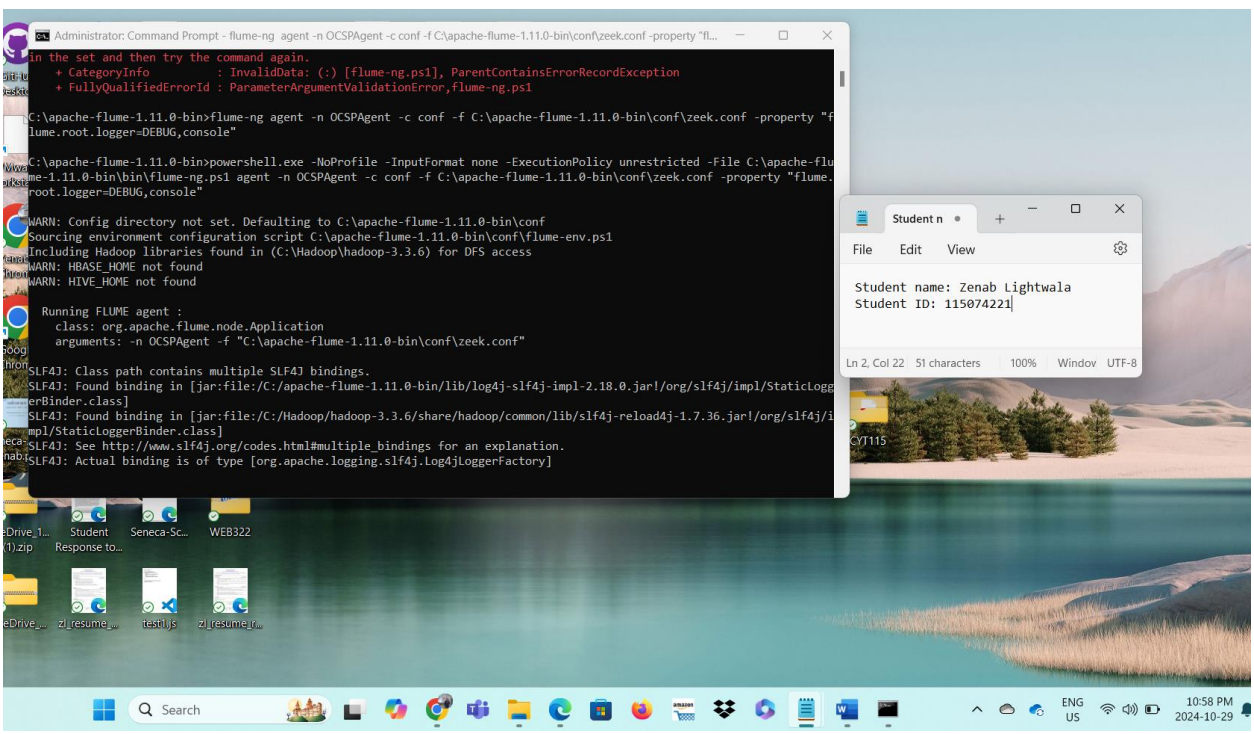
- Making a folder called zeek.conf in conf folder and making the changes with OCSPAgent



- Making a directory /user/Hadoop/zeek_logs/



- Confirming it on localhost:9870



- Running “stream_zeek.conf” in PowerShell with OCSP Agent. The command used flume-nd agent -n OCSPAgent -c conf -f C:\apache-flume-1.11.0-bin\conf\stream_zeek.conf -property “flume.root,logger=DEBUG.console”

SUMMARY:

Apache Flume is a system that works across many computers, is dependable, and can handle a lot of work. It's made to gather, combine, and move large amounts of data that are constantly coming in from different places to a central storage area, like the Hadoop Distributed File System (HDFS). Flume is especially good at managing data logs and events from sources like web servers, social media, and network traffic.

Capabilities and Features of Apache Flume-

Apache Flume has many features that make it a strong tool for bringing in data in big data projects. One of its main abilities is collecting data from many different sources. Flume can work with sources like files, log data, and even social media platforms. This flexibility helps users bring in data more smoothly and handle different types of data formats.

Another important feature of Flume is its reliability. It makes sure that data isn't lost during the process of collecting it, thanks to built-in features like acknowledgments and retries. When set up correctly, Flume can save data to a disk until it's successfully sent to its final destination, which is very important in situations where losing data is not an option.

Flume also handles large amounts of data efficiently and can be scaled up easily. You can add more agents to Flume as needed, so it can manage increasing amounts of data without slowing down. Plus, Flume offers different ways to store and send data (like memory, files, HDFS, and HBase), giving users the freedom to customize how data flows to meet their specific needs.

Additionally, Flume's ability to be expanded is a crucial feature. Developers can make custom parts, like sources, destinations, and channels, to fit particular needs. This ability to expand, along with its simple setup using an XML file, makes Flume a flexible tool for bringing in data.

How Apache Flume Works

Apache Flume's system is made up of different parts that work together to move data from where it starts to where it ends. The main parts in Flume are sources, channels, and sinks.

Sources: These are the parts of Flume that bring in data. A source could be a log file, a network connection, or any other data stream. Each source is designed to handle a particular type of data and format, making Flume flexible for various needs.

Channels: Channels act like a temporary storage area that holds the data before it gets sent to the sink. Flume offers different types of channels, such as memory channels and file channels, so users can pick the one that best fits their performance and reliability needs. Channels are important because they separate the sources from the sinks, ensuring data is safely stored until it can be processed.

Sinks: Sinks are the last part of Flume's process, where the data is sent to its final place. This could be a system like HDFS, HBase, or another storage solution. Sinks can be set up in different ways to make sure data is delivered quickly and reliably.

Flume's design also allows for multiple agents, which can operate on various machines within a cluster. This distributed setup enhances scalability and fault tolerance. In a standard setup, data moves from sources to channels and finally to sinks, ensuring a seamless and dependable data collection process.

Practical Outcomes of Using Apache Flume

In my practical use of Apache Flume, I concentrated on moving log files from the "Book of Zeek" into HDFS. The setup required configuring a Flume agent with particular sources, channels, and sinks designed for log data transfer. By using the flume-ng command, I managed to start the agent and keep an eye on the data transfer process.

The outcomes of this work were positive. Flume effectively gathered the log data, with very little data missing and a high rate of data transfer. The logging features gave immediate updates on the transfer process, helping me fix any problems quickly. Also, the option to expand the transfer process by adding more agents is a useful feature for managing bigger datasets in the future.

Overall, Apache Flume is a strong and adaptable tool for bringing data into big data systems. It can gather, combine, and move streaming data, and its well-designed structure helps it grow and stay dependable. This makes it a key part of the Hadoop system. Using Flume to bring in log files from "Book of Zeek" showed how good and reliable it is, proving its worth in today's data-focused world. As companies keep making lots of data, tools like Apache Flume will be very important for managing and processing that data well.